

AIN SHAMS UNIVERSITY-
FACULTY OF ENGINEERING



[MCT343]

Mechatronic System Design

PID Submission Report

Team 16

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Introduction

Precise motion control is fundamental to the performance of modern robotics and automation systems. This project centers on the implementation of a 5 Degree-of-Freedom (5-DOF) robotic arm, with a focus on controlling two of its joints driven by DC motors. To ensure accurate position and velocity regulation of these joints, two dedicated PID (Proportional-Integral-Derivative) controllers are designed and deployed, enabling closed-loop control for enhanced system stability and responsiveness.

Utilizing MATLAB/Simulink and embedded microcontroller programming, the control strategy is developed and refined through both simulation and real-time implementation. Each motor is governed by an independent PID loop—one for position and one for speed control—allowing for fine-tuned actuation under dynamic conditions. The tuning process considers key performance metrics including rise time, settling time, steady-state error, and robustness to disturbances.

After achieving satisfactory simulation results, the PID algorithms are implemented on a physical prototype using a microcontroller. The real-time control performance is then evaluated and compared with the simulated behavior to assess consistency and effectiveness. This integrated workflow bridges theoretical control concepts with practical applications in robotics.

Through the development of this 5-DOF robotic arm, the project provides hands-on experience in system modeling, PID controller design, dynamic simulation, and embedded system implementation—critical skills in the field of control systems and robotic engineering.

DC Motor Position Control Using PID with Encoder Feedback

To control the position of two DC motors using a PID controller based on encoder feedback, ensuring stability, minimal overshoot, and accurate steady-state response.

Experimental Procedure

➤ System Initialization

➤ Initial Testing

- Applied a step input (desired position) to the motors. Observed open-loop response: unstable, underdamped, high overshoot, and slow settling.

➤ Tuning Method: Trial and Error

- Started with all PID gains set to zero.
- Increased K_p incrementally:
 - System started to respond to input but was unstable and oscillated.
 - Increases speed of response.
 - High K_p reduces rise time but causes overshoot.
- Introduced K_i to eliminate steady-state error:
 - Response improved over time.
 - But too high K_i caused instability or integral windup.
 - Eliminates steady-state error.
- Added K_d to reduce oscillations:
 - Improves stability and damping.
 - Reduces overshoot and **oscillation**, but sensitive to noise.
- Add LPF filter:
 - Filters high-frequency noise in encoder signal to prevent noisy derivative action.

Final PID Parameters

The best performance (minimal overshoot, fast response, and zero steady-state error) was achieved with the following values:

- Underdamped response with acceptable overshoot (~5–10%).
- Settling time decreased significantly.
- Proper tuning of PID gains significantly improved control performance.
- The trial-and-error approach was effective but time-consuming.

Final Tuning Parameter

Tuning For Motor 1

Yellow : Actual position

Red : Target

Black: Control Action

- We designed the system with little overshoot for small friction in motion.

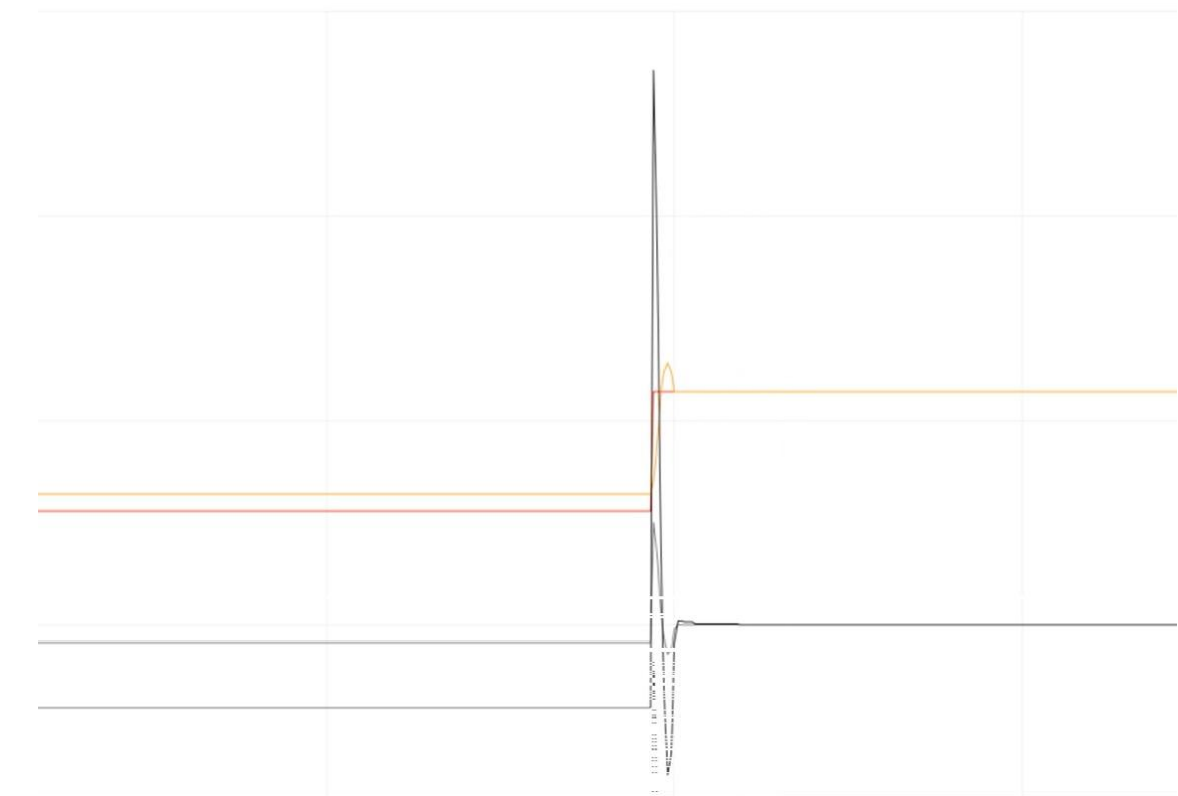


Figure 1 Response of Motor 1

	Kp	Ki	Kd
M1	5	0.4	0.3
M2	2.2	0	0.8

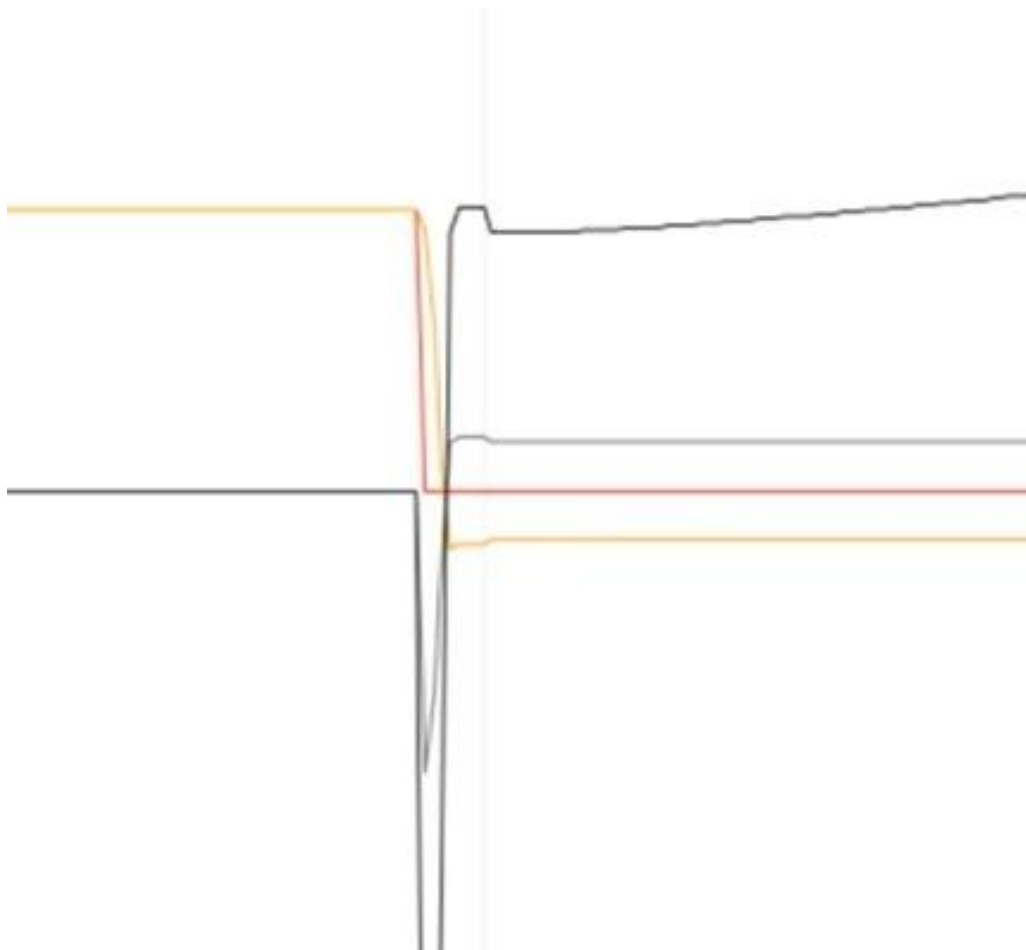


Figure 2 Response of Motor 1

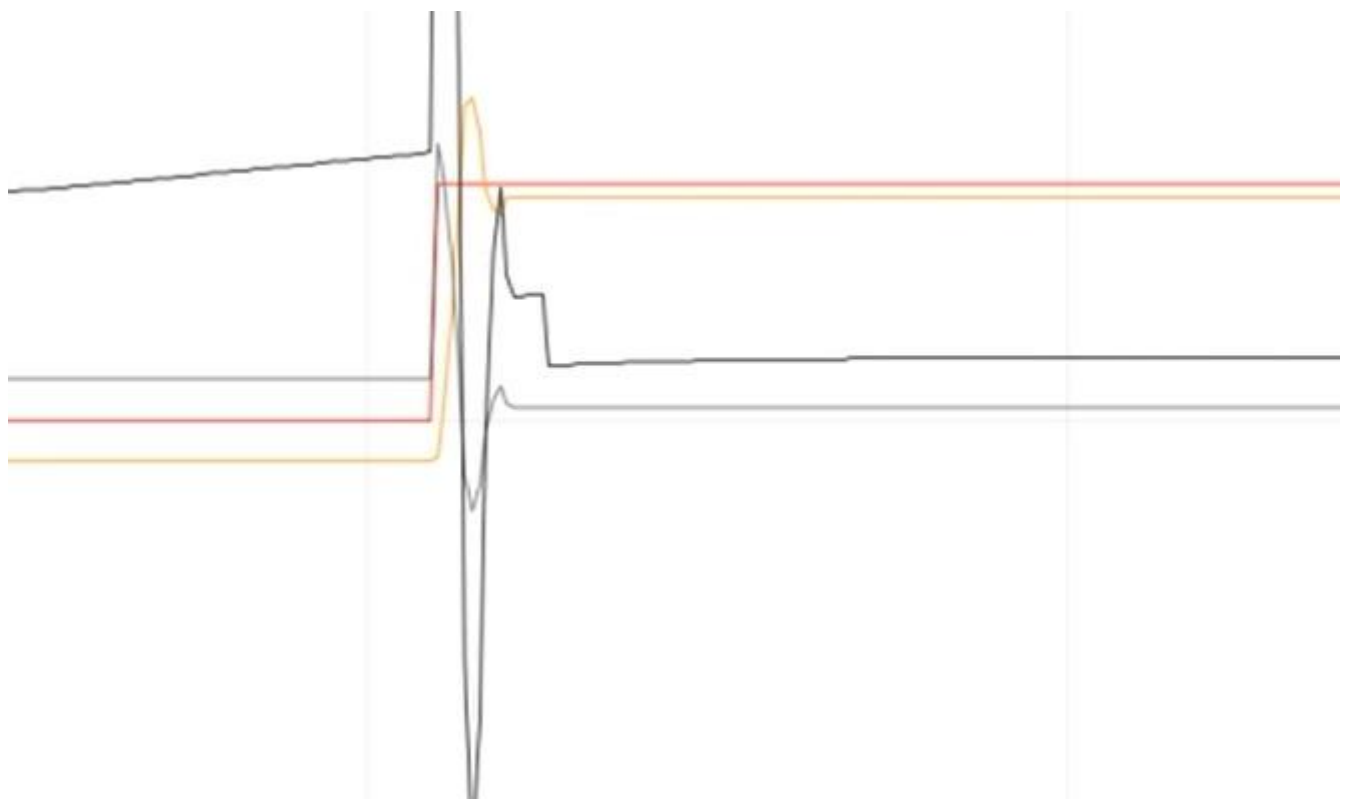


Figure 3 Response of Motor 1

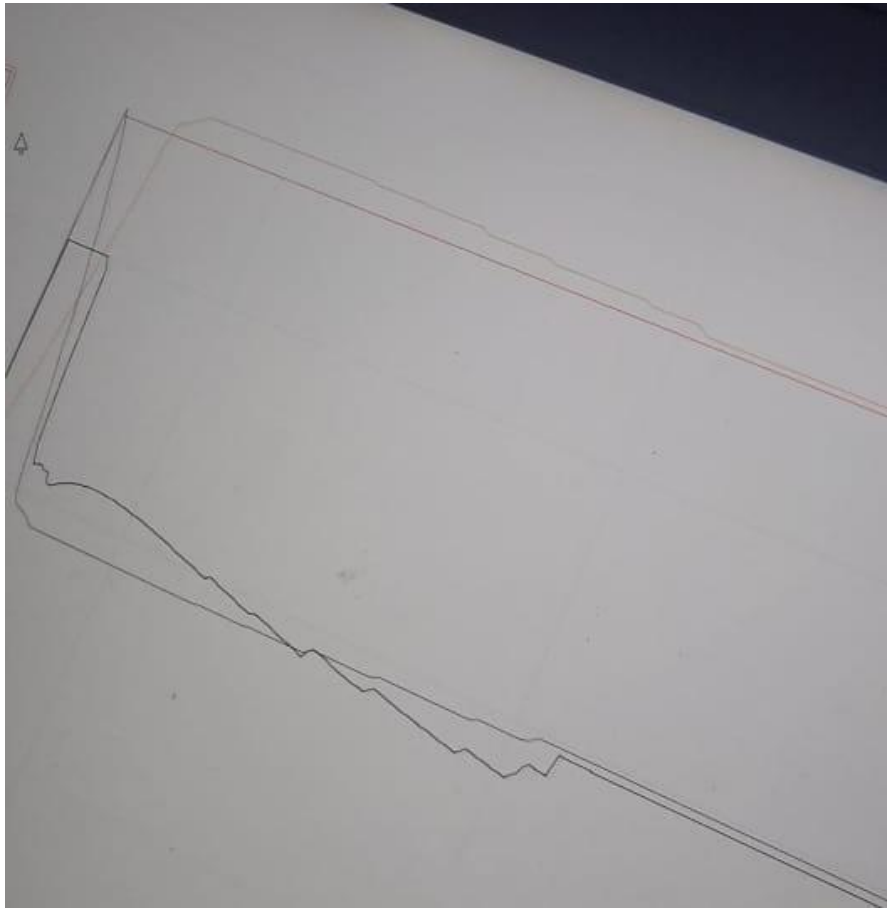


Figure 4 Response of Motor 2

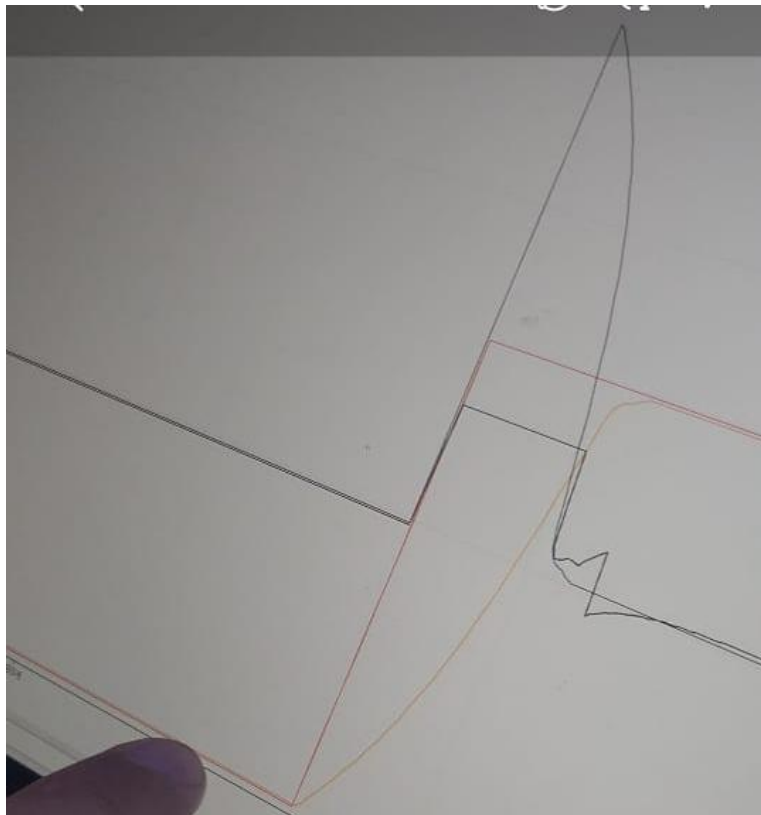


Figure 5 Response of Motor 2

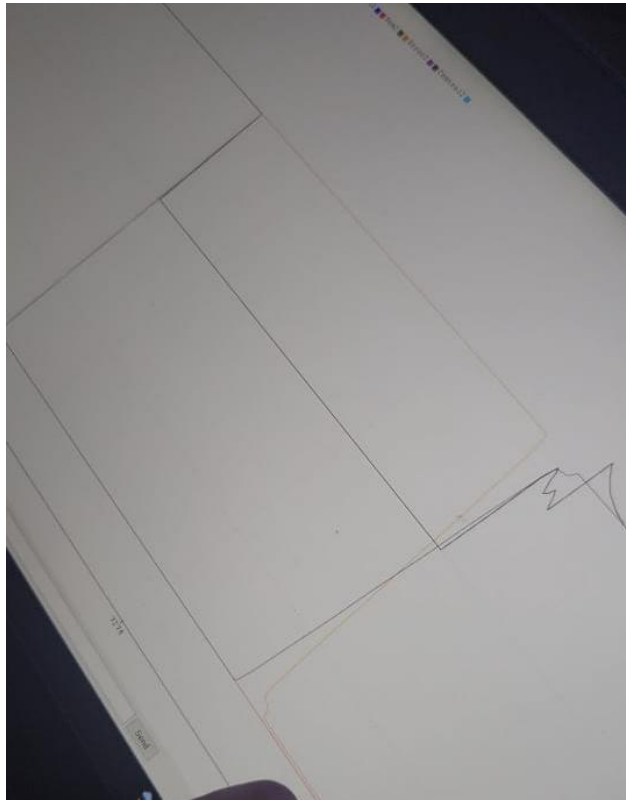


Figure 6 Response of Motor 2

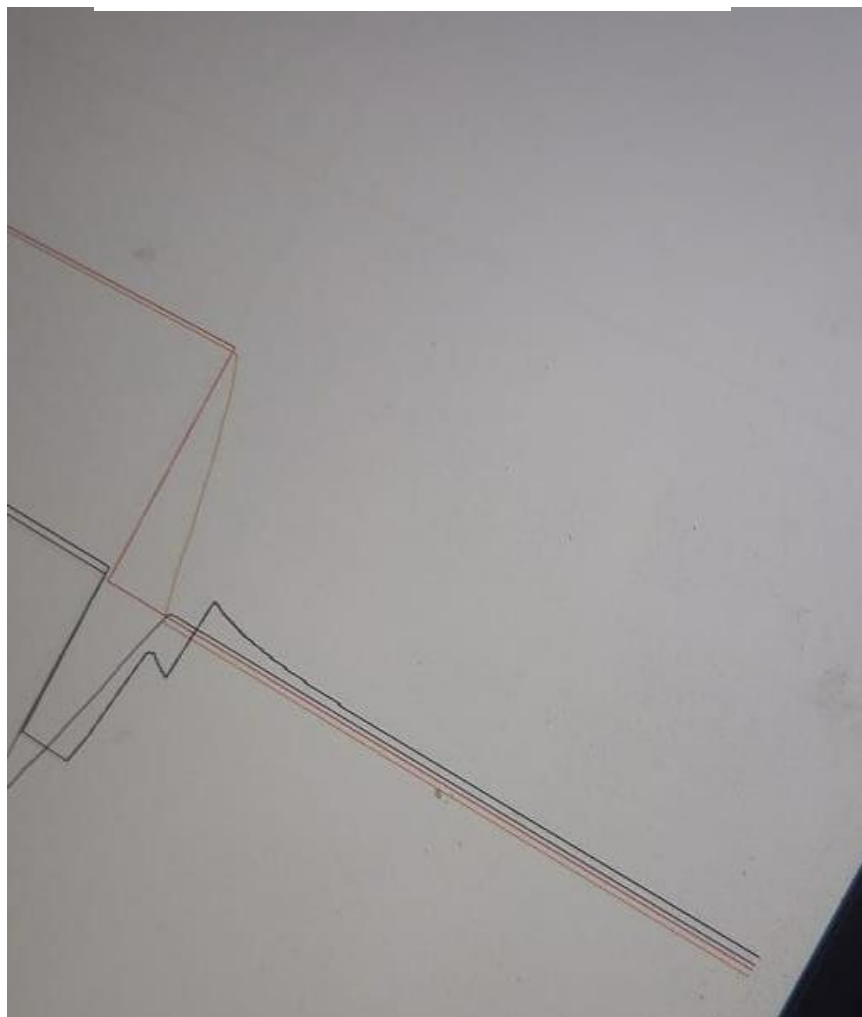


Figure 7 Response of Motor 2